Pre-Analysis Plan (PAP) – #ManyDaughters Project

Daniel Graeber & Carsten Schröder

2025-06-04

1 Overview and Objectives

This pre-analysis plan (PAP) specifies *ex-ante* how we will test the four causal hypotheses laid out by the #ManyDaughters organizers using the German Socio-Economic Panel (SOEP-Core). All modelling choices, variable definitions, exclusion rules and inferential criteria are fixed before accessing the full dataset, in line with project instructions.

2 Data and Sample

- Universe – SOEP-Core individual questionnaire (pl) merged with biography (biobirth, biogen), generated person (pgen), biography (biol) and household (hgen) files.
- Data structure – Hypothesis (1,4): cross-sectional, year t=2018; Hypotheses (2,3): unbalanced panel of adult respondents including all years where dependent and control variables are available.
- Baseline inclusion rules:

1. Individuals aged at least 18 and giving birth to at least one child during the observation period for dynamic models. For cross-sectional models, we will rely on samples of parents.
2. We will restrict the time wondow to observations from 2000 to 2019.
3. H2, H3: Individuals become at least one child during their participation in the sample. H1 & H4: Individuals that are parents. H1, H2, H4: both, single parents and couples; H3: Only opposite sex couples.
4. Completed individual questionnaire in that year (variable netto).
5. Only private households (variable pop)
6. Birth biography available in biobirth (to derive child sex composition).
7. Non-missing outcome, treatment and covariate information (see below).

- Key identifiers – pid (person), hid (household), syear (survey year) and pgpartnr (partner identifier) for household-level analyses.

3 Variables

3.1 Treatment – "Having at least one daughter"

Derived from KIDSEX[n], with n being the child order, in biobirth (1 male, 2 female will be coded as 1 being female and zero otherwise) and children's birth years.

- Indicator $Daughter_{it}$ = 1 if respondent has ≥1 living biological daughter by year t; 0 otherwise.

3.2 Controls

For data-limitation reasons outlined below, for H1 and H4 we will rely on cross-sectional models, for H2 and H3 we will rely on panel models.

For our cross-sectional analysis of Hypothesis 1 and 4, we will rely on (conditional) independence, i.e., that conditional on having children, the gender of the child is independent of the vector of potential outcomes. To put it differently, conditional on having a child, the gender of the child is as good as random. While this is not testable directly, we can test one implication: That the observed characteristics of the parents of male and female children are equal (in expectation). To do so, we will assess the balance of a wide range of pre-determined or exogenous characteristics of the parents depicted in Table 1. Balance evaluation will rely on standardized differencesusing Stata's "stddiff" package suited for continuous and categorial variables. We will consider any standardized difference that is larger than 20 in magnitude as problematic. In case of an imbalances in covariates, i.e., standardized differences larger than 20, we will implement augmented inverse probability weighting, i.e., Stata's teffect aipw, for hypothesis 1 and 4.

Besides these control variables, we will also include the number of children as derived from biobirth.dta in the specification.

For the dynamic models, the choice of control variables is guided by the following considerations:

1.) Include all characteristics that jointly determine children's gender as well as the outcome of interest.
2.) Do not include characteristics that are determined by the treatment and the outcome or another (un)observed characteristic to avoid collider bias.

Hence, our set of control variables focuses on mostly pre-determined individual characteristics (usually defined before having a child), such as education, migration background, or cognitive ability. If time invariant, they will be absorbed by individual fixed effects. Consequently, the only control variables relevant for our dynamic panel models are:

- Year fixed effects
- Individual fixed effects
- Number of children (only non-predetermined or exogenous)

Table 1: Variables for covariate balancing and control variables in H1 and H4

Covariate	SOEP dataset / file	Key variable(s)	Operationalisation (coding / transformation)
Respondent sex	ppathl – longitudinal person-tracking file	sex (1 = male, 2 = female)	Binary dummy *Male* = 1, *Female* = 0
Year of birth / cohort	ppathl	gebjahr	Continous for balancing test, categorical in Equation (1)
State of birth	ppathl – longitudinal person-tracking file	birthregion_ew (federal-state code)	Indicator *East* = 1 if birthregion_ew = 21, 0 otherwise (includes -2 to accompany migrants but excludes -1)
Migration background	ppathl	migback (1 = none, 2 = first, 3= second generation)	Keep categorical variable as is
Father's highest education	bioparen	fsedu	Keep as categorical variable as is (drop -5, -4, -2, and -1)
Mother's highest education	bioparen	msedu	Keep as categorical variable as is (drop -5, -4, -2, and -1)
Grew up with single-parent	biol	lb0066	Dummy = 1 if lb0066 <15, 0 otherwise (excluding negative values)
Respondent highest school degree	pgen	pgisced11	Keep as is, exclude -1 but keep -2.
Age at first interview	ppathl	syear_first – gebjahr, with syear_first being first year with a completed person interview	Continuous years
Partner's school degree	partner's pgen (merge)	pgisced11_partner, where phisced11_partner is the pgisced11 of the partner	Same collapse as respondent

3.3. Outcomes

The following table provides the outcome variable for each hypothesis.

#	Hypothesis & construct	SOEP variable(s) (long file pl unless noted)	Coding & transformation
1	**H1 – Support for gender equality (attitudes)**	plh0308_v2 – *"Best if man and woman work the same amount so they can share the responsibility."* 7-point item (1 = "strongly disagree", 7 = "strongly agree") (paneldata.org); from 2018 only	Keep original 1–7 scale; standardized to have mean zero and standard deviation in the control group
2	**H2 – Left vs. right political orientation (binary)**	plh0012_h "Party you feel closest to" (paneldata.org)	leftparty = 1 if party ∈ {Die Linke, Greens, SPD, Die Piratenpartei}; 0 if party ∈ {FDP, CDU/CSU, AfD, DVU, NPD, Republikaner}; If respondents indicate first and second vote, we will refer to the second vote.
3	**H3 – Gender equality in household behaviour**	pli0043_h – hours **per weekday** respondent spends on housework ("Hausarbeit Std./Werktag") (paneldata.org) … plus the *same variable for the co-resident partner* (merge on partner PID)	1. $Hours_{week}$ = pli0043_h 2. Compute **share**: $\boldsymbol{Share_{it}} = \frac{\boldsymbol{Hours_{week}}}{\boldsymbol{Hours_{week}+Hours_{week,partner}}}$ 3. Measure of equality $\lvert 0.5 - \boldsymbol{Share_{it}} \rvert$ **.** The higher this value is, the larger the inequality in housework.
4	**H4 – Gender differences in support for gender equality (attitudes)**	plh0308_v2 – *"Best if man and woman work the same amount so they can share the responsibility."* 7-point item (1 = "strongly disagree", 7 = "strongly agree") (paneldata.org) in 2018	Keep original 1–7 scale; standardized to have mean zero and standard deviation in the control group

4 Empirical Strategy

Depending on the year(s) in which outcomes are available, we will run cross-sectional or dynamic panel models.

The SOEP provides self-reported attitudes on gender equality in survey year 2018 only. Hence, our estimation will be cross-sectional for Hypotheses (1) and (4). For Hypotheses (2) and (3), we will estimate dynamic models.

Hypotheses 1 & 4

Hypothesis 1

For Hypothesis (1), our empirical model takes the form

$$Y_i = \beta_0 + \beta_1 Daughter_i + \boldsymbol{\beta_2 Children_i} + \boldsymbol{\gamma' X_i} + \varepsilon_i. \qquad (1)$$

In Equation (1), we regress the outcome, e.g., gender equality attitudes measured in 2018, on an indicator that is equal to one if the person has at least one child and zero otherwise, a row vector with indicators for the number of children, $\boldsymbol{Children_i}$, as derived from

biobirth.dta, a vector of controls, as listed in Table 1, $\boldsymbol{X_i}$. In Equation (1), ε_i is the potentially non-i.i.d. error term. Therefore, we estimate a heteroscedasticity robust variance-covariance matrix. The estimate of β_1 then reflects the effect of having at least one daughter compared to having only sons or no children.

We test Hypothesis (1) via the following one-sided hypothesis test:

$$H0: \beta_1 = 0 \qquad v.s. \qquad H1: \beta_1 > 0.$$

Hypothesis 4

For Hypothesis (4), we include an additional gender-interaction to estimate the differential effect for fathers compared to mothers. Hence, the regression model takes the form

$$Y_i = \alpha_0 + \alpha_1 Father_i + \alpha_2 Father_i * Daughter_i + \boldsymbol{\alpha_3} Father_i * \boldsymbol{Children_i} + \boldsymbol{\delta}' Father_i * \boldsymbol{X_i} + \alpha_5 Daughter_i + \boldsymbol{\alpha_6 Children_i} + \boldsymbol{\phi}' \boldsymbol{X_i} + \mu_i, \qquad (2)$$

Where $Father_i$ is an indicator that is one if the parent is male and zero otherwise. The estimate of α_2 corresponds to the differential effect of having at least one daughter, compared to having only sons or no children, for male parents, with female parents being the reference group. Again, we estimate heteroscedasticity robust variance covariance matrices.

We test Hypothesis (4) via the following one-sided hypothesis test:

$$H0: \alpha_2 = 0 \qquad v.s. \qquad H1: \alpha_2 > 0.$$

Hypotheses 2 & 3

For our dynamic models, we rely on difference-in-difference models. We test for the existence of pre-trends. We will rely on two way fixed effects models from hdidregress in Stata. Our control group will comprise only not-yet-treated individuals, i.e., individuals that did not yet give birth to a daughter but will do so at a later point in time. If common trend assumptions do not hold, we will also apply augmented inverse probability weighting (hdidregress aipw in Stata). For this, we would rely on our set of predetermined and exogenous variables (Table 1). We consider 3 pre- and 4 post-treatment periods. We will report aggregated treatment effects with associated clustered standard errors and significance stars based on p-values from a z-statistic.

Hypothesis 2

To test Hypothesis (2), we estimate the following empirical difference-in-differences model:

$$Y_{it} = \zeta_0 + \zeta_1 Daughter_i + \boldsymbol{\zeta_2 Children_i} + \boldsymbol{\kappa Z_i} + \boldsymbol{\chi Syear_t} + \rho_i. \qquad (3)$$

For this, we regress out outcome on the indicator for having at least one daughter, a vector of indicators for the number of children, a vector of indicators for the years since birth, a vector of indicators for each individual, $\boldsymbol{Z_i}$, and a vector of indicators for each survey year, $\boldsymbol{Syear_t}$. We estimate variance covariance matrices, clustered on the individual level. The

estimate of ζ_1 corresponds to the effect of having at least one daughter, compared to having only sons or no children, on our outcome.

We test Hypothesis (2) via the following one-sided hypothesis test:

$$H0: \zeta_1 = 0 \qquad v.s. \qquad H1: \zeta_1 > 0.$$

Hypothesis3

To test Hypothesis (3), we estimate the following empirical difference-in-differences model:

$$Y_{it} = \eta_o + \eta_1 Daughter_i + \boldsymbol{\eta_2 Children_i} + \boldsymbol{\lambda Z_i} + \boldsymbol{\nu Syear_t} + \psi_i. \quad (4)$$

For this, we regress out outcome on the indicator for having at least one daughter, a vector of indicators for the number of children, a vector of indicators for the years since birth, a vector of indicators for each individual, $\boldsymbol{Z_i}$, and a vector of indicators for each survey year, $\boldsymbol{Syear_t}$. We estimate variance covariance matrices, clustered on the individual level. The estimate of η_1 corresponds to the effect of having at least one daughter, compared to having only sons or no children, on our outcome.

We test Hypothesis (3) via the following one-sided hypothesis test:

$$H0: \eta_1 = 0 \qquad v.s. \qquad H1: \eta_1 < 0.$$

7 Multiple Testing

The many-daughters project builds around one pre-specified primary hypothesis (daughters matter). However, because the projects asks for testing four hypothesis simultaneously, there is an increased risk of false positives (Type I errors). Hence, we will also report Bonferroni adjusted p-values.